AN ABELIAN CATEGORY HAS ALL FINITE LIMITS AND COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for finite limits, each ingredient being either a defining property of abelian categories or an identification displayed in the span; the closing sentence 'The argument for finite colimits is similar but dual to this.' is the author's own routine duality remark, preserved verbatim.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Finite limits from products and equalizers. Categories, Lemma categories-lemma-finite-limits-exist. Supplies the reduction of the existence of all finite limits to the existence of finite products and of equalizers.
- Equalizers as kernels. Same chapter, the description of the equalizer of two parallel morphisms as the kernel of their difference. Supplies the identification that makes the equalizer exist in an abelian category.

Source reference. Original lemma 002O: sources/stacks-categories/pinned-original-source.tex, source lines 2265–2265. Navigation pointer invoked at proof line 910; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. All finite limits and finite colimits exist in $\mathcal{A}$.*

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. To show that finite limits exist it suffices to show that finite products and equalizers exist, see Categories, Lemma 002O. Finite products exist by definition and the equalizer of $a, b : x \to y$ is the kernel of $a - b$. The argument for finite colimits is similar but dual to this. □